

Ma Siu Zheng

Malaysia | 011-33528500 | siuzheng18@gmail.com | github.com/SiuZheng

Summary

AI/ML Engineer with practical experience building and deploying real-time Computer Vision and LLM-powered systems. Skilled in PyTorch, YOLO, SAM2, LLM finetuning, OCR, and deep learning workflows, with hands-on expertise in constructing FastAPI backends, REST pipelines, and GPU-accelerated inference services. Experienced in deploying AI applications using Gradio, Streamlit, Dify, and Chatwoot integrations, and developing Android AI features using Kotlin. Adept in object detection, segmentation, feature engineering, and delivering end-to-end AI solutions from model training to production deployment.

Experience

Dah Reply Sdn Bhd | AI Backend Developer Intern

Oct 2024 – Jan 2025

- Built and deployed AI backend systems using **FastAPI**, including a real-time **SAM2 segmentation server** with streaming support.
- Created a **Gradio UI** for webcam/image/video segmentation and object selection via image reference or text description.
- Developed an **AI chatbot** by scraping company website data and connecting it to WhatsApp via **Chatwoot**, **Typebot**, and **Dify**.
- Integrated automated reply pipelines through backend routing and data structuring.

Projects

Lung Cancer Detection (Deep Learning, Medical AI)

- Built a CT-scan lung cancer classification system using a custom Hybrid Convolutional-Transformer-Like Feature Network (HCTLFN) combined with AGOC preprocessing. Achieved ~91% accuracy.
- **Tech:** Python, PyTorch, EfficientNetV2, BiGRU, Feature Engineering (AGOC)

AI Storybook Generator (LLM + Image Generation)

- Developed an AI pipeline that generates children's storybooks with consistent characters and narratives for a kindergarten in Ipoh. Integrated LLMs for storytelling and image generation workflows.
- **Achievement:** transformed a one-month manual process into a one-day automated workflow, enabling the organization to avoid hiring additional personnel and achieve a 50% reduction in operational costs.
- **Tech:** Python, LLMs, LangChain, Streamlit

Malaysia Car Plate Detection (Computer Vision)

- Implemented a car plate detection and OCR system using a finetuned YOLO11n and EasyOCR. Designed preprocessing steps for night/day performance to enhance recognition accuracy.
- **Tech:** Python, YOLO11n (finetuned), EasyOCR, Gradio, Data Preprocessing

AI Study – Smart Learning Assistant (Android + AI)

- Built an Android application that generates personalized study plans, flashcards, quizzes, and explanations using AI. Integrated backend AI processing via FastAPI and Dify.
- **Tech:** Kotlin, Android, FastAPI, AI Integration, Dify



FYP — Disease Prediction & Health Chatbot (Android + ML)

- Created a health prediction app using XGBoost models paired with an AI chatbot to answer health-related questions. Included data preprocessing and backend API support.
- **Tech:** Kotlin, Android, Machine Learning, FastAPI, Dify

Education

Universiti Tunku Abdul Rahman (UTAR) – Kampar

2025

Bachelor of Computer Science – CGPA 3.38

Skills & abilities

- **AI/ML:** PyTorch, YOLO, SAM2, LLMs, OCR, Model Finetuning (YOLO, CNN, LLM), Machine Learning, Data Preprocessing, Deep Learning, AI Agent, RAG
- **Backend:** FastAPI, REST APIs, LangChain
- **Deployment:** Gradio, Streamlit, GPU Inference, Dify, Chatwoot Integration
- **Android:** Kotlin, Jetpack, API & AI integration
- **Computer Vision:** Object Detection, Segmentation, Feature Engineering
- **Tools:** Git, Docker (basic), Typebot, GCP, ComfyUI, LM Studio, Unsloth